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AI for Bharat Hackathon

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Team Name : Robots

Team Leader Name : Ujjwal Godara

Problem Statement : Track 1 - AI for Retail, Commerce & Market Intelligence

MandiMitra: Agentic Market Intelligence Copilot

An AI-powered multi-agent system built on Amazon Bedrock that empowers **150M+ Indian farmers** with real-time mandi price intelligence, sell/hold recommendations, negotiation support, and agricultural knowledge — all through **conversational Hindi interface**.

The Problem:

- India has 7,000+ APMC mandis generating massive price data. Yet 86% of small farmers sell at whatever price the local arthiya offers — losing 15-30% of potential crop value due to information asymmetry.
- Current tools (Agmarknet, eNAM, Kisan Suvidha) are English-only dashboards with no contextual intelligence.

Our Solution:

- MandiMitra uses Amazon Bedrock Multi-Agent Collaboration (SUPERVISOR mode) to orchestrate 5 specialist AI agents — Price Intelligence, Sell Advisory, Negotiation Prep, Weather, and Knowledge Base (RAG) — that query real-time data via 13 tool functions.
- Farmers ask in Hindi and get data-driven answers with price comparisons across 1,451 mandis, optimal sell timing, and shareable negotiation briefs.
- Conversational Hindi Interface for real-time insights, data-driven answers, price comparison, sell/hold advice.

Live Prototype: https://d2mtfau3fvs243.cloudfront.net | **GitHub:** github.com/ujjwalgodara9/AI-For-Bharat--AWS

Why AI is Required

- Raw price data from 7,000+ mandis is meaningless without intelligent analysis.
- Multi-agent AI converts raw prices into SELL/HOLD/SPLIT decisions.
- Combines price trends, shelf life, MSP benchmarks, weather forecasts, and transport costs.
- Hindi voice queries make it accessible to 150M+ farmers with limited literacy.

How AWS Services are Used (12 services)

- Amazon Bedrock Multi-Agent Collaboration (Supervisor + 4 sub-agents, Claude Sonnet 4)
- Bedrock Knowledge Base + OpenSearch Serverless + Titan Embeddings v2 (RAG)
- DynamoDB (16,500+ records)
- Lambda (3 functions)
- API Gateway
- EventBridge (daily cron)
- CloudFront + S3 (HTTPS CDN)
- Bedrock Guardrails v3
- CloudWatch (7 alarms)
- LangFuse.

What Value AI Adds

- First agentic AI for Indian agri-commerce combining real-time mandi data + weather-price correlation + transport-cost-adjusted net realization.
- Knowledge Base (MSP, crop calendar, storage, mandi guide).
- Multiiingual conversational interface.
- Not a dashboard — a personal market intelligence advisor that speaks Hindi.
- Recovers Rs.50,000-1,00,000 per farmer annually.

Key Features of MandiMitra

Price & Advisory

Real-Time Price Intelligence: Live prices for 20+ commodities across 1,451 mandis in 24 states with trend analysis (rising/falling/stable) and MSP comparison.

Smart Sell Advisory: AI-powered SELL/HOLD/SPLIT recommendations factoring 14 variables (price trends, perishability, storage costs, weather, MSP, season).

GPS Best Mandi Finder: Transport-cost-adjusted net realization comparison across mandis within 50-100km using 60+ GPS coordinates.

Negotiation Brice Brief: Shareable one-page brief with MSP reference, local/best price, trend direction, and fair price estimate.

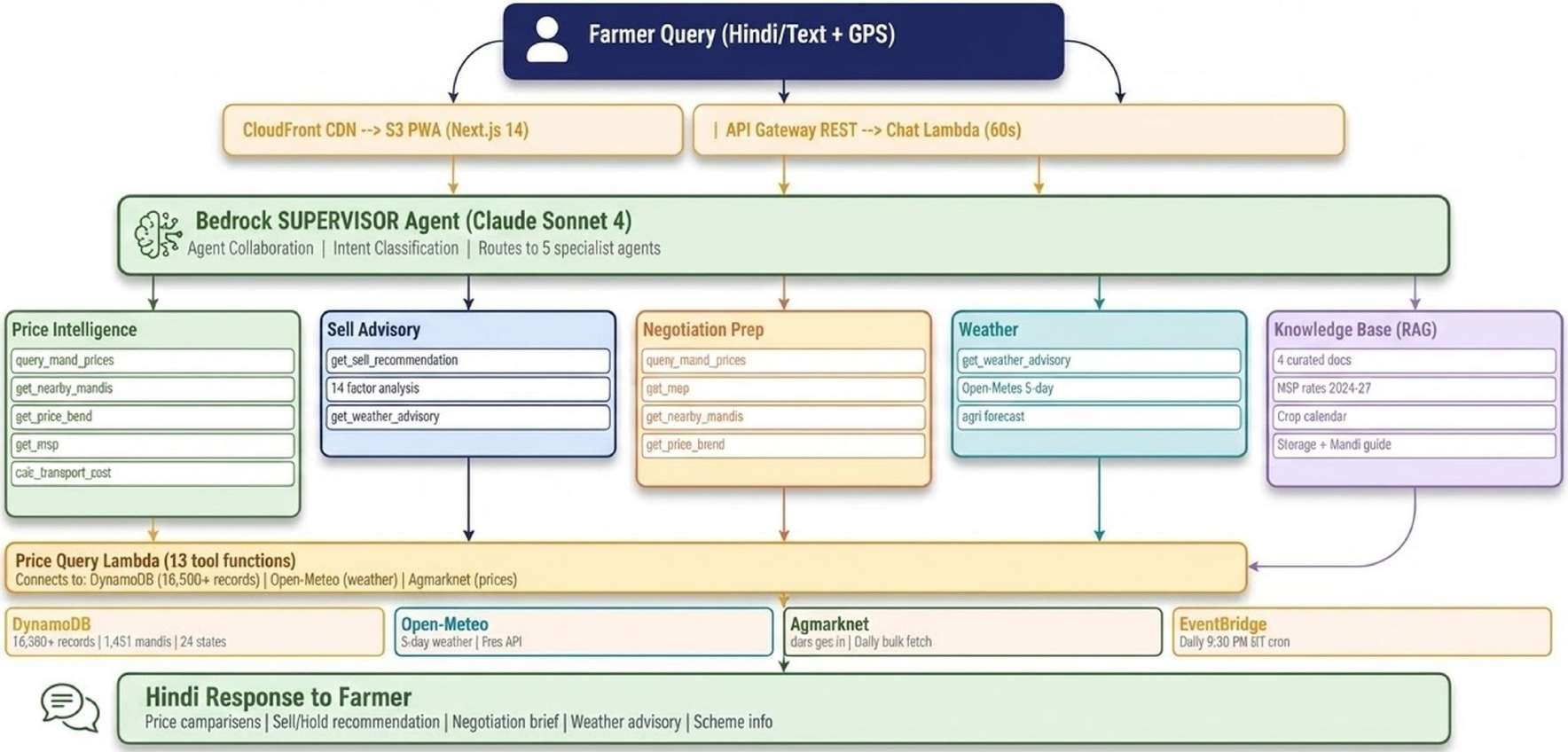
Accessibility & Tech

Hindi Voice Interface: WhatsApp-style chat with Web Speech API voice input in Hindi/Hinglish for low-literacy users.

5-Day Weather Advisory: Agricultural forecast with sell-timing guidance linking rain/temperature to crop storage impact via Open-Meteo.

Knowledge Base (RAG): Instant answers on MSP rates (2024-27), crop calendars, post-harvest storage, mandi procedures, e-NAM via 4 curated docs.

Multi-Agent Orchestration: Complex queries auto-routed across 5 specialist agents + 13 tool functions for comprehensive, multi-step answers.


Farmer Query (Hindi/Text + GPS)
CloudFront CDN --> S3 PWA (Next.js 14)
| API Gateway REST --> Chat Lambda (60s)
Bedrock SUPERVISOR Agent (Claude Sonnet 4)
Agent Collaboration | Intent Classification | Routes to 5 specialist agents
Price Intelligence
query_mand_prices
get_nearby_mandis
get_price_bend
get_msp
cale_transport_cost
Sell Advisory
get_sell_recommendation
14 factor analysis
get_weather_advisory
Negotiation Prep
queny_mand_prices
get_mep
get_nearby_mandis
get_priee_brend
Weather
get_weather_advisory
Open-Metes S-day
agri forecast
Knowledge Base (RAG)
4 curated docs
MSP rates 2024-27
Crop calendar
Storage + Mandi guide
Price Query Lambda (13 tool functions)
Connects to: DynamoDB (16,500+ records) | Open-Meteo (weather) | Agmarknet (prices)
DynamoDB
16,380+ records | 1,451 mandis | 24 states
Open-Meteo
S-day weather | Fres API
Agmarknet
dars ges in | Daily bulk fetch
EventBridge
Dally 9;30 PM ßIT cron
Hindi Response to Farmer
Price camparisens | Sell/Hold recommendation | Negotiation brief | Weather advisory | Scheme info

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AI for Bharat Hackathon

USERS: Farmer (Hindi/English) | FPO Manager | Trader | Extension Worker

FRONTEND: Next.js 14 (SSR + PWA) | Tailwind CSS | Web Speech API (Hindi STT/TTS) | Geolocation API (GPS) | CloudFront HTTPS CDN

COMPUTE: API Gateway REST | Lambda: Chat | Lambda: Price Query | Lambda: Ingestion | EventBridge | S3 Hosting

AMAZON BEDROCK — AGENTIC AI LAYER

Supervisor Agent (Claude Sonnet 4)
SUPERVISOR Mode | Intent Classification | Routes queries

PriceIntelligence
5 tools: prices, mandis, trends, MSP, transport

SellAdvisory
14 factors SELL/HOLD/SPLIT + weather

Negotiation
4 tools: prices, MSP, mandis, trends

Weather
Open-Meteo 5-day agri forecast

KB (RAG)
4 docs: MSP, calendar, storage, mandi

Bedrock Knowledge Base (RAG)
Titan Embeddings v2 | FlexFl dim | OpenSearch Serverless (FAISS HNSW)
4 docs: MSP Rates 2024-27 | Crop Calendar | Post Harvest Storage | Mandi Procedures

Bedrock Guardrails (v3 LIVE)
Content filtering | Safety rules | PII protection | Factual grounding

Price Query Lambda — 13 Tool Functions
query_mandi_prices | get_nearby_mandis | get_price_trend | get_msp | calculate_transport_cost | get_sell_recommendation | get_weather_advisory

DATA LAYER:

DynamoDB
16,300+ records, 3 GSIs, 222 commodities

S3
Frontend + KB docs + audit logs

OpenSearch Serverless
FAISS vector store for RAG

Agmarknet/data.gov.in
2000+ mandis, daily batch

Open-Meteo
Free 5-day weather API

Monitoring: CloudWatch (7 recorls) | LangFuse (LLM tracing | tracing) | CloudFront access logs

Snapshots of the prototype:

1. Welcome Screen – Hindi landing page with MandiMitra branding, GPS location picker, and 6 quick-action buttons (Live Price, Best Mandi, Negotiate, Sell/Hold, MSP Info, Weather)

2. Price Check – Ask in Hindi e.g. "MP mein gehun ka bhav" → 10+ mandis with modal/min/max prices, MSP comparison, 7-day trend arrows, SVG sparkline price charts

3. GPS Best Mandi – Share location → mandis within 50km ranked by net realization (price minus transport cost), with distance and cost estimates

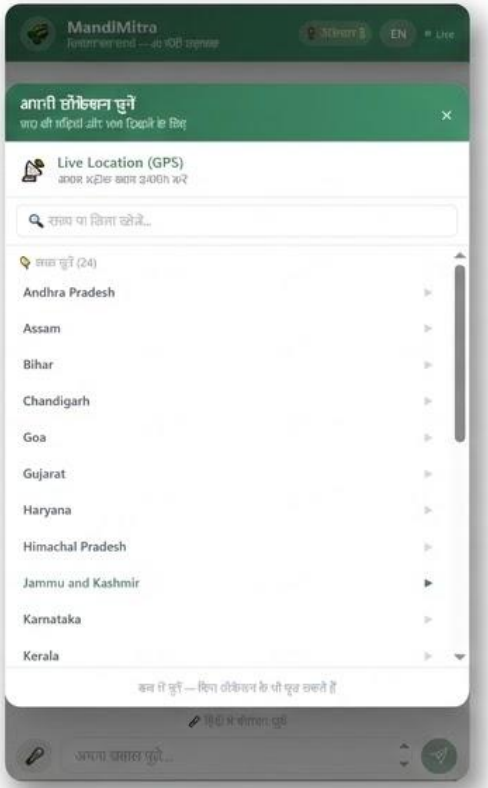
4. Sell Advisory – AI analyzes 14 factors (price trend, perishability, MSP, weather, season, storage) → SELL/HOLD/SPLIT recommendation with Hindi reasoning

5. Negotiation Brief – Formatted price brief with MSP reference, local vs best nearby mandi price, trend direction — shareable via WhatsApp

6. Weather Advisory – 5-day agricultural forecast from Open-Meteo with sell-timing guidance linking weather to crop storage impact

7. Knowledge Base – Ask "Wheat ka MSP?" → Official 2024-27 rates with YoY change, source citation via RAG over 4 curated documents

Test it yourself : https://d2mtfau3fvs243.cloudfront.net



Location Selection & Mandi Discovery

Home Screen & Core Features

Conversational Price Inquiry & Results

Technology Stack

AI/ML: Amazon Bedrock Agents (multi-agent with tool use, Claude Sonnet 4), Bedrock Knowledge Bases (RAG, 4 curated docs), Titan Embeddings v2 (1024-dim vector search), OpenSearch Serverless (FAISS HNSW vector Bedrock Guardrails v3 (content safety, PII protection)

Compute: AWS Lambda (3 functions — Chat Handler 60s, Price Query 30s with 13 tools, Data Ingestion 900s), API Gateway REST (CORS + rate limiting), CloudFront HTTPS CDN (enables Voice + GPS APIs on mobile)

Dev & Ops: GitHub (version control), CloudWatch (7 alarms — errors, throttles, duration), LangFuse (LLM tracing & observability)

Data: Amazon DynamoDB (16,500+ price records, 3 GSIs, PAY_PER_REQUEST, 222 commodities), Amazon S3 (static hosting + KB docs + audit logs), EventBridge (daily 9:30 PM IST automated ingestion cron)

Frontend: Next.js 14 (SSR + static export + PWA installable), Tailwind CSS (mobile-first responsive), Web Speech API (Hindi/English STT + TTS), Geolocation API (GPS-based mandi discovery)

External Data: Agmarknet/data.gov.in (commodity prices, 2000+ mandis), Open-Meteo (free 5-day weather forecast), Nominatim/OSM (reverse geocoding), Ministry of Agriculture (MSP reference rates 2024-27)

Total: 12 AWS services actively used in production

Estimated Monthly Cost (1,000 DAU MVP)

Service Breakdown

- **Amazon Bedrock** (Claude Sonnet 4 — 50K queries/month): ~Rs.8,000
- **DynamoDB** (on-demand, 16GB, 500K reads/month): ~Rs.800
- **Bedrock Knowledge Base** (OpenSearch Serverless 2 OCU): ~Rs.3,000
- **Lambda** (100K invocations, 256MB, avg 3s): ~Rs.200
- **S3 + CloudFront CDN** (10GB storage + HTTPS): ~Rs.200
- **API Gateway + EventBridge** (REST API + daily cron): ~Rs.300
- **CloudWatch** (7 alarms + Lambda monitoring + logs): ~Rs.300

Total & Projections

TOTAL: ~Rs.12,800/month (~$150/month)

- Scales to **~Rs.50,000/month** at 10,000 DAU.
- **100% serverless** — zero idle cost, pay-per-use pricing.
- Current prototype phase cost: **~Rs.1,000/month**.

Prototype Performance report/Benchmarking:

Response Time

Average end-to-end response: 15-21 seconds (multi-agent reasoning + DynamoDB + tool execution). Price Query Lambda: ~2-3 seconds. Chat Handler Lambda timeout: 60 seconds.

Data Scale

16,500+ DynamoDB price records (growing daily) | **222** commodities | **24** states | **1,451** mandis | **421** districts

Reliability & Monitoring

7 CloudWatch Alarms (errors, throttles, duration) — all OK. Bedrock Guardrails v3 filtering harmful content. LangFuse LLM tracing for full observability on every invocation. Daily automated data ingestion via EventBridge (9:30 PM IST).

Infrastructure: 100% serverless — auto-scales from 0 to millions with zero infrastructure management. No fixed costs. 12 AWS services in production.

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Additional Details/Future Development (if any):

TARGET IMPACT: 150M+ small farming households lose 15-30% of crop value to information asymmetry. MandiMitra democratizes market intelligence, potentially recovering Rs.50,000-1,00,000 per farmer annually.

SCALABILITY: 100% serverless (Lambda + DynamoDB + Bedrock). Adding new mandis, commodities, and languages requires only data pipeline config — zero architectural changes. Auto-scales to millions of users.

BUSINESS MODEL: Free for individual farmers (funded by FPO/institutional subscriptions). Premium tier for FPOs, aggregators, and commodity traders with bulk analytics and API access.

GOVT ALIGNMENT:
Complements eNAM (digital mandi), PM-KISAN, PMFBY, and Doubling Farmers Income mission.

ROADMAP:
WhatsApp Bot integration | Multi-language (Marathi, Telugu, Tamil, Gujarati, Punjabi) | ML-based 7-day price prediction | Push notifications for price alerts | Direct buyer-seller marketplace | Crop photo quality grading | Satellite data (NDVI) for yield-adjusted forecasting

Prototype Assets

GitHub Public Repository:

https://github.com/ujjwalgodara9/AI-For-Bharat--AWS

Demo Video (Max 3 Minutes):

Drive Link

Live Prototype URL:

https://d2mtfau3fvs243.cloudfront.net

Project Summary: MandiMitra — AI-powered multi-agent market intelligence copilot for Indian farmers. 5 Bedrock Agents, 13 tool functions, 16,500+ price records, 1,451 mandis, Hindi voice interface, 100% serverless on 12 AWS services.

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Thank You